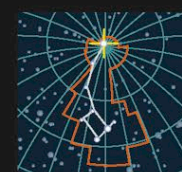


DOUBLE STAR

Polaris



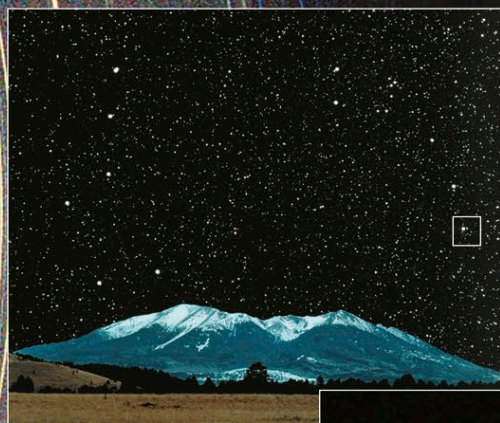
DISTANCE FROM SUN
430 light-years

MAGNITUDE 2

SPECTRAL TYPE F7

URSA MINOR

Polaris is famous as the current north Pole Star and consequently is known to every observer of the northern sky (see panel, below). However, it is also an interesting system in terms of its component stars. Polaris is a double star consisting of Polaris A, a supergiant, and Polaris B, a main-sequence star. The two stars can be separated through a modest amateur telescope, and Polaris B was first detected by William Herschel in 1780. The distance between them has been estimated at more than 190 billion miles (300 billion km). Polaris A is more than 1,800 times more luminous than the Sun and is also a Cepheid variable with a period of just under 4 days (see p.282). The radial velocity, or line-of-sight motion, of Polaris has been accurately measured (see p.70) and found to vary regularly with a period of 30.5 years. This indicates that Polaris is also an astrometric binary—that is, the presence of an unseen companion is detected by the movement it induces in the primary star (see p.274). The companion, which was seen for the first time in a Hubble Space Telescope image in 2005, orbits Polaris with a 30.5-year period but is so faint that it has no effect on Polaris's spectrum.



DISTINCTIVE STAR

One of the best-known stars in the northern sky, Polaris lies just away from the celestial pole, in the tail of Ursa Minor, the Little Bear. This telescope view reveals the faint companion, Polaris B, but a second smaller companion remains invisible.

EXPLORING SPACE

CELESTIAL SIGNPOST

Polaris has long been regarded as the most important star in the northern sky. Because it is located almost directly overhead at the north pole, it has long been used, just like a compass, to locate north. By calculating the relative angle of Polaris above the horizon, travelers by land and sea have also used Polaris to establish approximate latitudinal positions on the Earth's surface. The status accorded to Polaris by disparate cultures is reflected in their myths. In Norse mythology, Polaris was the jewel on the head of the spike that the gods stuck through the universe. The Mongols called Polaris the Golden Peg that held the world together. In ancient China, Polaris was known as Tou Mu, the goddess of the North Star.



IN THE LITTLE BEAR'S TAIL

In Arabic mythology, Polaris was an evil star who killed the great warrior of the sky. The dead warrior was said to lie in the tail of the little bear, a constellation that also represented a funeral bier.